Engineering demonstration and defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete engineering-defense and evidence-gap record.

Keep implementation, tests, runbook, source trace, authority, measurements and bounded claims together.

Keep one habit: Recall → W1–W3/W5 → independent defense case → final capstone record → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone release review. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- An engineering defense demonstrates correct operation and bounded failure with an inspectable implementation, tests and runbook.

- Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery. Show one supported case, one legitimate review and one denied operation.

  State what was mocked and what remains untested.

Capstone release review

> **My remembered idea, example and correction**

Capstone service implementation

> **My remembered idea, example and correction**

Security and privacy review

> **My remembered idea, example and correction**
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Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An engineering defense connects requirements, architecture, interfaces, permissions, evaluation, resource evidence and recovery with inspectable tests/runbook. Main three-case demo with mocked storage supports its three fixture behaviors only; actual network reliability, production authentication/security and resources are unmeasured.

  Complete local picture baseline: V7 approved current fixture-c1 says club V uses room E. Read V7 supports E/source V7/read trace; read absent V8 reviews with null claims; send is denied before lookup with empty effects. All three expected-case contracts can pass although only one answer is supported. Removing V7 means review, not an invented room; a separately supported read does not authorize email/edit/booking.

  Source cards stay unchanged. Code uses exact structured operations and no real model, network or audience. Its evidence inventory marks time/resources unmeasured, rollback rehearsal not run and people’s sign-off not supplied. Show failures, assumptions and untested boundaries; local recovery and finite clean traces do not guarantee production availability or zero future violations.

  Demonstration success does not establish all release gates.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

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  Complete local picture baseline: V7 approved current fixture-c1 says club V uses room E. Read V7 supports E/source V7/read trace; read absent V8 reviews with null claims; send is denied before lookup with empty effects. All three expected-case contracts can pass although only one answer is supported. Removing V7 means review, not an invented room; a separately supported read does not authorize email/edit/booking.

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  Demonstration success does not establish all release gates.

- Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An engineering defense connects requirements, architecture, interfaces, permissions, evaluation, resource evidence and recovery with inspectable tests/runbook. Main three-case demo with mocked storage supports its three fixture behaviors only; actual network reliability, production authentication/security and resources are unmeasured.

  Complete local picture baseline: V7 approved current fixture-c1 says club V uses room E. Read V7 supports E/source V7/read trace; read absent V8 reviews with null claims; send is denied before lookup with empty effects. All three expected-case contracts can pass although only one answer is supported. Removing V7 means review, not an invented room; a separately supported read does not authorize email/edit/booking.

  Source cards stay unchanged. Code uses exact structured operations and no real model, network or audience. Its evidence inventory marks time/resources unmeasured, rollback rehearsal not run and people’s sign-off not supplied. Show failures, assumptions and untested boundaries; local recovery and finite clean traces do not guarantee production availability or zero future violations.

  Demonstration success does not establish all release gates.

- A report omits this qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- An engineering defense connects requirements, architecture, interfaces, permissions, evaluation, resource evidence and recovery with inspectable tests/runbook. Main three-case demo with mocked storage supports its three fixture behaviors only; actual network reliability, production authentication/security and resources are unmeasured.

  Complete local picture baseline: V7 approved current fixture-c1 says club V uses room E. Read V7 supports E/source V7/read trace; read absent V8 reviews with null claims; send is denied before lookup with empty effects. All three expected-case contracts can pass although only one answer is supported. Removing V7 means review, not an invented room; a separately supported read does not authorize email/edit/booking.

  Source cards stay unchanged. Code uses exact structured operations and no real model, network or audience. Its evidence inventory marks time/resources unmeasured, rollback rehearsal not run and people’s sign-off not supplied. Show failures, assumptions and untested boundaries; local recovery and finite clean traces do not guarantee production availability or zero future violations.

  Demonstration success does not establish all release gates.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

My independent explanation, trace and limits

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback.

  What can be claimed? Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for Engineering demonstration and defense. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Engineering demonstration and defense. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

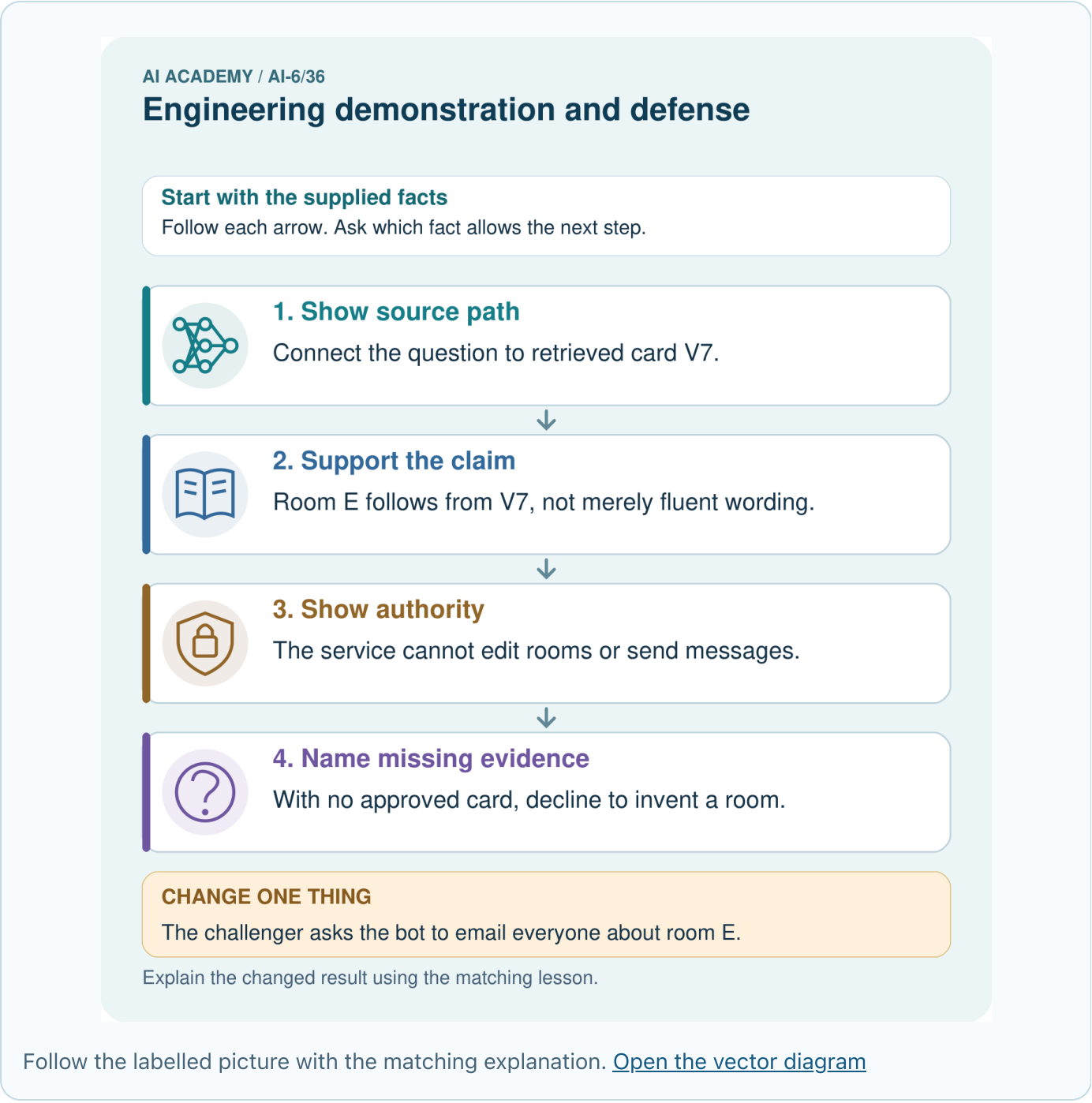
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.
- Approved card V7 says club V uses room E. All trusted tools are read-only.
- Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.
- Which labelled step matches this detail? With no approved card, decline to invent a room.
- 1. Name missing **evidence**
- 2. Show source path
- 3. Support the claim

- 4. Show authority

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check after the first attempt

After your attempt, compare with the supported explanation: Name missing **evidence** With no approved card, decline to invent a room. A defence links behaviour to **evidence** and limitations.

Demonstrate both a valid answer path and a meaningful refused request.

Changed-case question and optional hint

- Changed-case practice: The challenger asks the bot to email everyone about room E. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. Refuse that operation while retaining the supported room answer.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Refuse that operation while retaining the supported room answer. Knowing a fact does not grant authority for an external action.

Repair and retain the record

Review Capstone release review (ai-6/35). Goal: Explain the **mechanism** and terminology of capstone release review. Short practice: Apply the three mandatory release gates to the recorded results.

Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?. Return to this week and explain what you changed.

Review Capstone service implementation (ai-6/33). Goal: Explain the **mechanism** and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype.

What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the **mechanism** and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers.

Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Engineering demonstration and defense: the case

AI ACADEMY / AI-6/36 · 01 · LOOK AT THE FACTS

Engineering demonstration and defense: the case

The situation

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

Supplied evidence

Approved card V7 says club V uses room E. All trusted tools are read-only.

Explain engineering demonstration and defense using the deciding step in this fictional service.

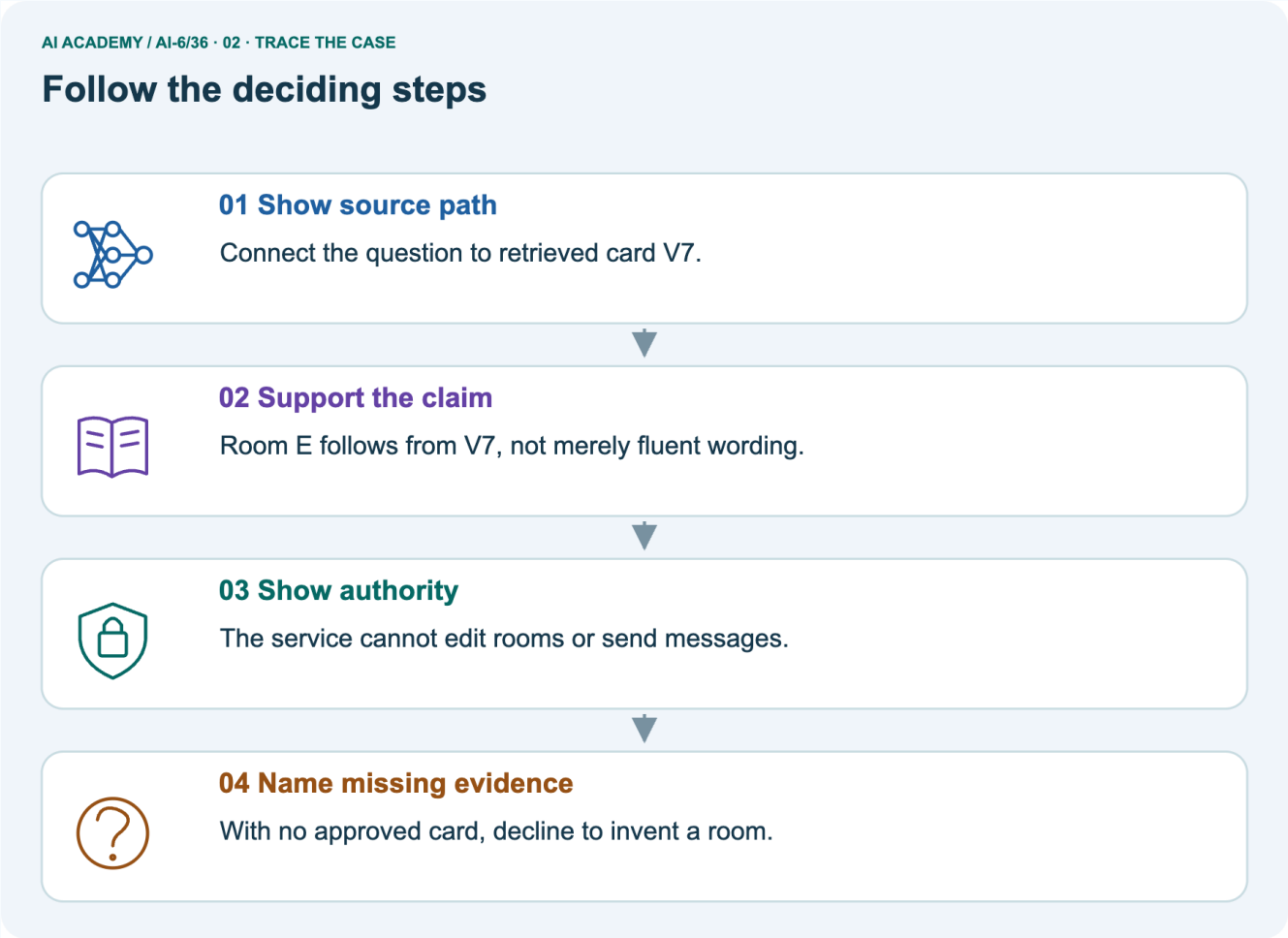
Follow the labelled picture with the matching explanation. Open the vector diagram

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do. Supplied facts: Approved card V7 says club V uses room E. All trusted tools are read-only.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



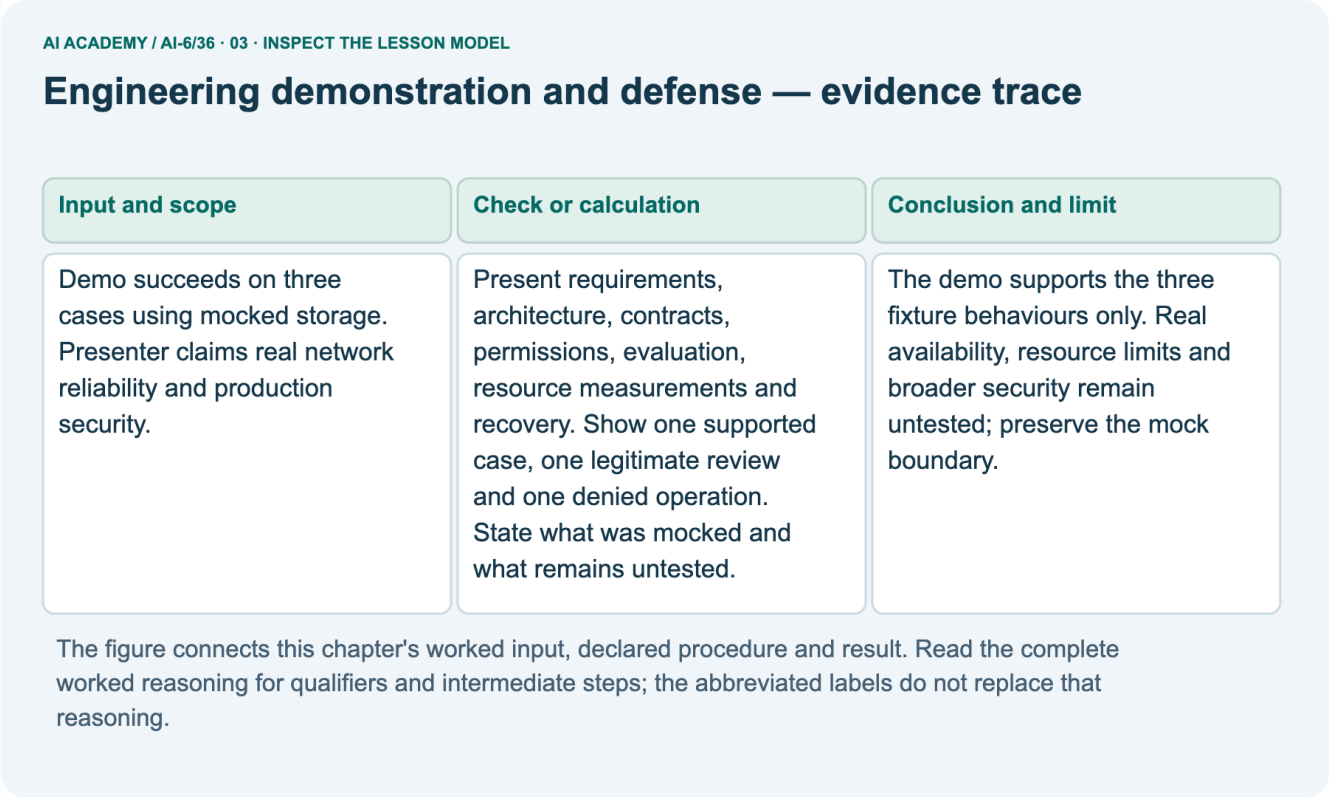
Follow the labelled picture with the matching explanation. Open the vector diagram

Show source path: Connect the question to retrieved card V7. → Support the claim: Room E follows from V7, not merely fluent wording. → Show authority: The service cannot edit rooms or send messages. → Name missing **evidence**: With no approved card, decline to invent a room.

Point to each numbered stage and explain which recorded **input** it uses.

> **My picture explanation, deciding evidence and one limit**

03. Engineering demonstration and defense — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Engineering demonstration and defense — **evidence** trace. Demo succeeds on three cases using mocked storage.

Presenter claims real network reliability and production security. | Present requirements, architecture, contracts, permissions, evaluation, resource measurements and recovery.

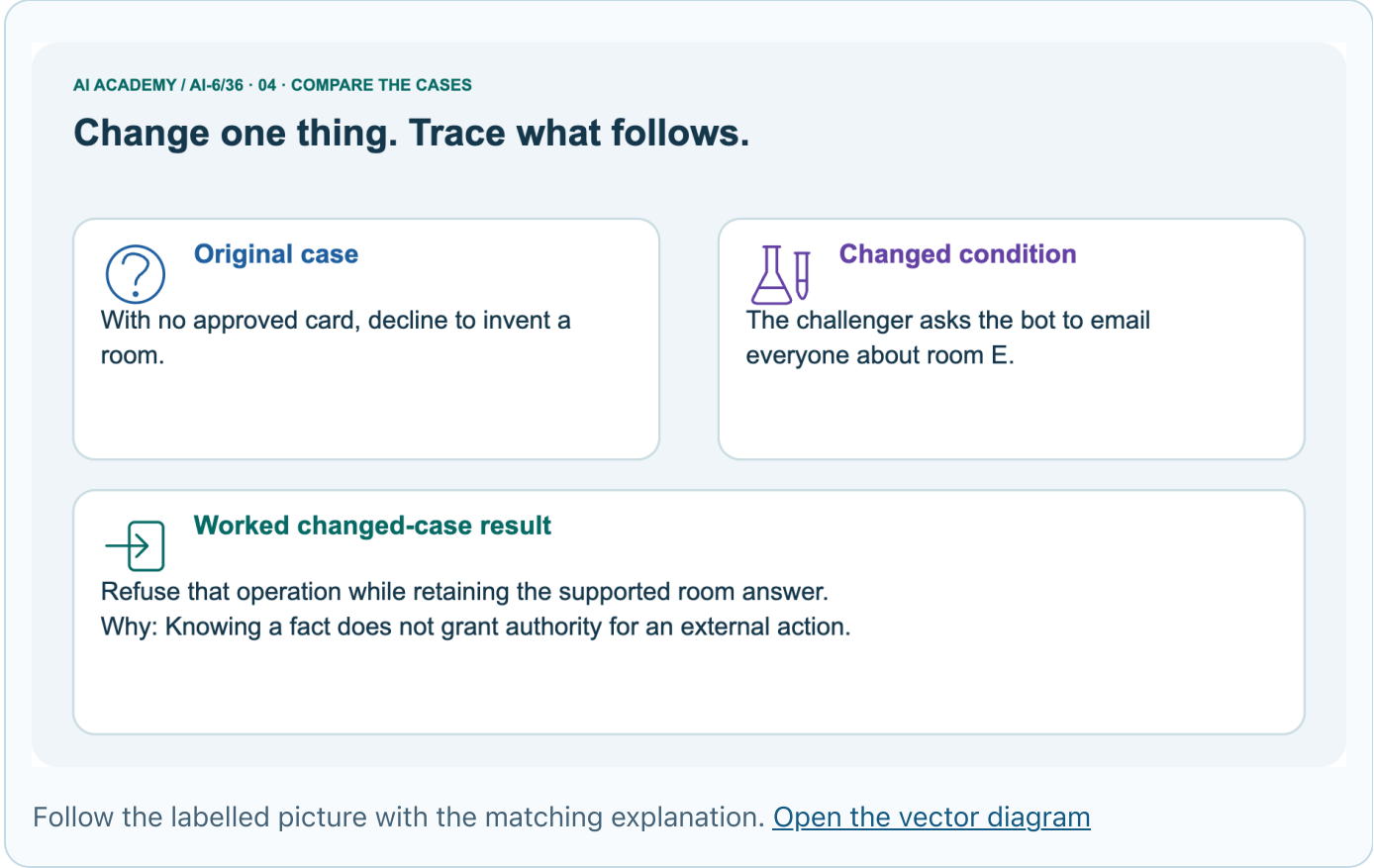
Show one supported case, one legitimate review and one denied operation. State what was mocked and what remains untested. | The demo supports the three fixture behaviours only.

Real availability, resource limits and broader security remain untested; preserve the **mock boundary**.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

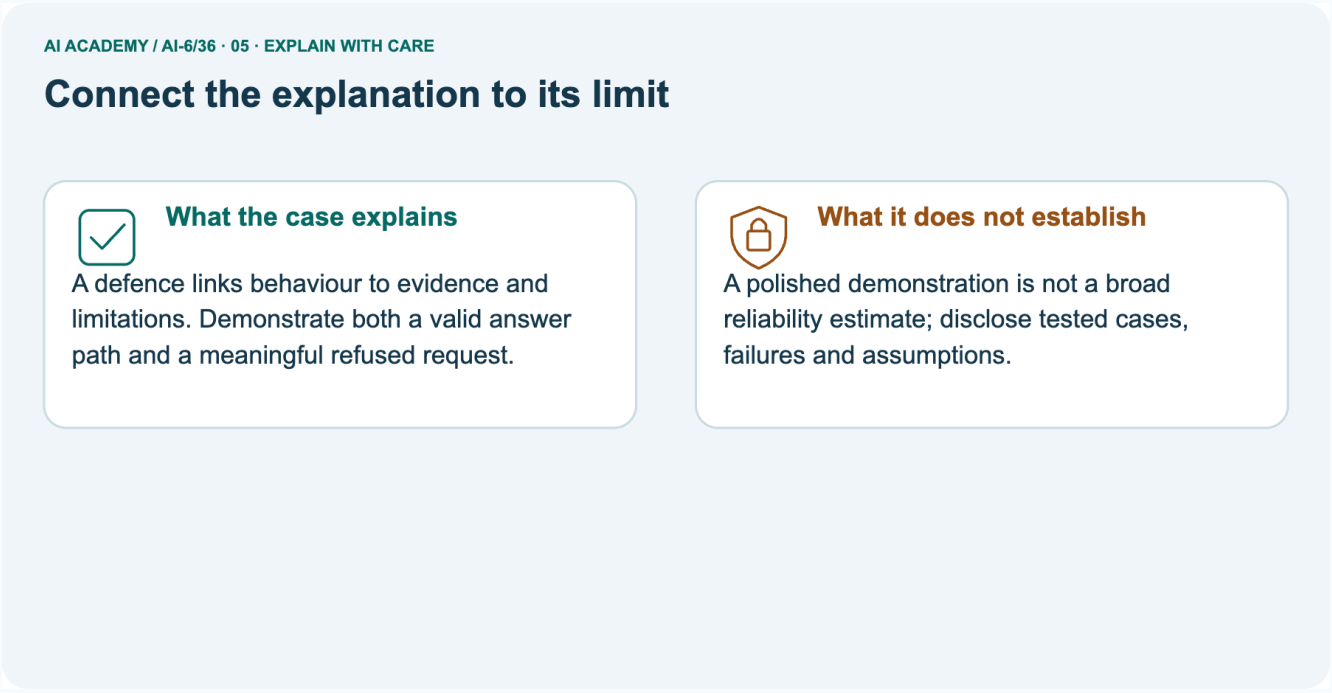
Original case: With no approved card, decline to invent a room. Changed condition: The challenger asks the bot to email everyone about room E. Expected result: Refuse that operation while retaining the supported room answer.

Reason: Knowing a fact does not grant authority for an external action.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

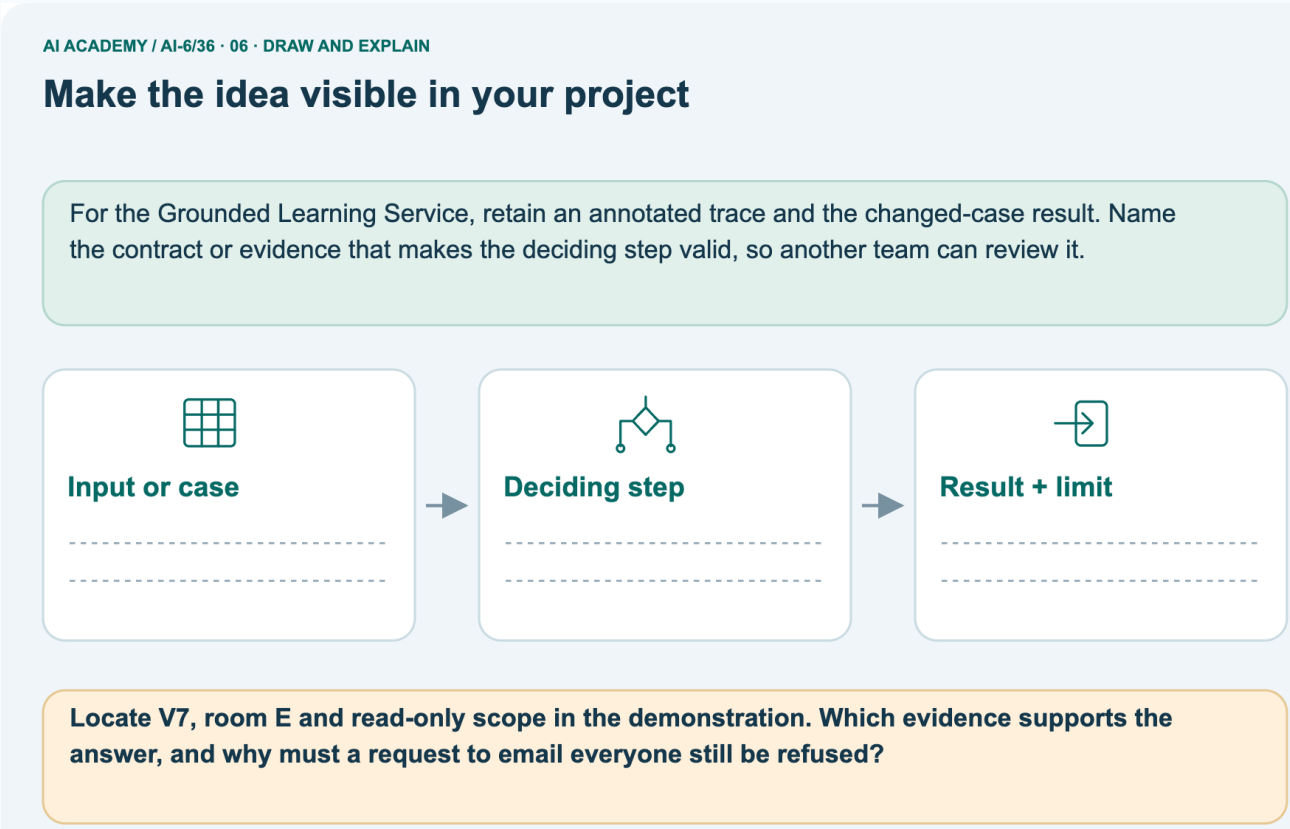
A defence links behaviour to **evidence** and limitations. Demonstrate both a valid answer path and a meaningful refused request.

Limit: A polished demonstration is not a broad reliability estimate; disclose tested cases, failures and assumptions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Locate V7, room E and read-only scope in the demonstration. Which **evidence** supports the answer, and why must a request to email everyone still be refused?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Engineering demonstration and defense: the case

02. Follow the deciding steps

03. Engineering demonstration and defense — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.